Option B – Medical Physics

Question				Marking details	AO1	AO2	AO3	Total	Maths	Prac
8	(a)			Two graphs one completely above the other labelled high voltage with a skewed normal distribution (1) Line spectrum in the same place on both graphs (1) Minimum wavelengths on both graphs labelled and not at (0,0) (1) Minimum wavelengths given as 4.125×10^{-11} and 2.5×10^{-11} [m] (1)	1 1 1	1		4	4	
	(b)	(i)		Rearrangement: $v = \frac{\Delta f c}{2 f \cos\theta}$ (1) v = 0.12 [m s^{-1}] (1)		2		2	2	
		(ii)		Monitor thinning arteries / clots / foetus (development) etc Accept high heart rate		1		1		
	(c)	(i)		Hydrogen nuclei / protons precess about B-field (1) Absorbing radio waves causes flip / change orientation (1) Flipping back (of nuclei precession) emits radio waves (1)	3			3		
		(ii)		Frequency = 5.96×10^{7} [Hz] (1) Wavelength = 5.0[3] [m] (1) Deduct 1 mark for power of 10 errors		2		2	2	
	(d)	(i)		**Either** $2 \times m_e \times 931 = 2 \times$ gamma energy **OR** $m_e \times 931 =$ gamma energy **OR** in numbers **OR** $0.000549 \times 1.67 \times 10^{-27} \times c^2$ (1) Correct answer = 0.511 MeV **OR** 8.20×10^{-14} J (1) Award 1 mark only for answer of 1.02 Deduct 1 mark for power of 10 errors		2		2	2	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(ii)		Expensive / needs a cyclotron		1		1		
	(e)			X-rays are ionizing damage the foetus (1) Fluoroscopy is also ionizing (1) B scan (ultrasound) is ideal / good (1) A scan only gives the depth of the foetus (in the womb) (1) CT scans are ionizing and would damage the foetus (1) Penalise once only for harmful radiation			5	5		
				Question 8 total	**6**	**9**	**5**	**20**	**10**	**0**